# ARCHIT AGRAWAL

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## PROFESSIONAL EXPERIENCE

**EdPlus, Arizona State University** **06/2026 - Present**
*AI Software Engineer* *Phoenix, AZ*

- Own architecture and delivery of an AI survey analytics platform on AWS Bedrock and Step Functions, with cited answers and deterministic calculations.
- Built a survey analytics application with a Nuxt frontend and NestJS backend, sharing TypeScript contracts across 70 API routes.
- Reduced AWS pipeline runtime from 500 to 85 seconds by replacing two managed-agent stages with direct tool calls.
- Cut incorrect answers 85% in an evaluation against 164 published survey figures by adding answer verification before display.
- Validated 25 statistical tests and six causal estimators against SciPy and statsmodels; achieved 68% on StatQA test selection.
- Improved accuracy from 60% to 93% on the hardest benchmark questions by consolidating column definitions in a shared semantic layer.
- Increased recovery of reference headline findings from 13% to 52–59% on held-out surveys using LLM-selected analyses and deterministic calculations.
- Prevented mislabeled and dropped records with schema-constrained output and row correlation checks in the data-processing pipeline.

**MyStage Music Inc.** **07/2025 - 06/2026**
*Founding AI/ML Engineer* *Remote*

- Consolidated a 14-Cloud-Function backend into a checkpointed Python/LangGraph workflow on GCP Cloud Run, reducing production service dependencies 64%.
- Built FastAPI and Algolia search with sub-50 ms latency, real-time Firestore synchronization, 200+ pytest tests and Logfire tracing.
- Scaled ingestion to 70,000+ records daily from 650+ venues using Playwright, transactional locking and entity resolution.
- Implemented checkpointed pause-and-resume so records missing context wait for additional information without failing the ingestion batch.
- Made ingestion retries safe with deterministic task IDs and atomic create-if-absent writes, preventing duplicate jobs from overwriting in-flight work.

**EdPlus, Arizona State University** **09/2023 - 05/2025**
*Software Engineer* *Phoenix, AZ*

- Built a multi-tenant course assistant from React UI to retrieval services, used by 1,000+ faculty creating courses for 60,000+ students.
- Established Prompt Flow release checks for groundedness, relevance and coherence, evaluating prompt changes against measured results before deployment.
- Implemented tenant isolation in retrieval services and configuration-based onboarding for additional colleges.
- Added hybrid keyword and vector retrieval to resolve exact course codes alongside semantic document search.
- Reduced transcript-analysis time from four hours to 15 minutes using a Neo4j knowledge graph and validated Cypher queries.

**Arizona State University** **08/2024 - 05/2025**
*Student Researcher* *Phoenix, AZ*

- Built a PyTorch and Diffusers benchmark for text-to-video models that scores physical plausibility separately from visual quality, holding seed, resolution and sampler fixed.
- Found visual quality barely predicts physical plausibility: the top model by human preference ranked last on causal consistency.

**Knowledge Exchange for Resilience, Arizona State University** **06/2024 - 08/2024**
*Data Research Aide* *Phoenix, AZ*

- Reduced p95 retrieval latency 60% at 95% recall using hybrid FAISS and Neo4j search.
- Built a FastAPI and pgvector collaborator recommender achieving 85% top-k precision in evaluation.

**Zeus Learning** **01/2022 - 07/2023**
*Software Engineer | Intern Jan–Jun 2022; engineer Jun 2022–Jul 2023* *Mumbai, India*

- Reduced deployment time 70% and production incidents 40% through containerized CI/CD and SonarQube quality gates.
- Migrated a C#/.NET monolith to Dockerized Kubernetes microservices, reducing infrastructure cost 20% and resource footprint 35%.

## PROJECTS

**SRP Electric MCP Server** | GitHub **12/2025 - 01/2026**

- Built a Python/FastMCP integration exposing seven utility-account tools for billing, usage and outage queries, with an interactive setup wizard.
- Reverse-engineered the portal's login flow and API calls, since the utility publishes no API documentation.

**MCP GitHub PR Review Agent** **07/2025 - 08/2025**

- Built a TypeScript PR-review integration combining code diffs, tickets and CI results into schema-validated inline findings.
- Made review comments idempotent across pushes and synchronized ticket context with Asana.

## TECHNICAL SKILLS

**Languages & APIs:** Python, TypeScript, SQL, FastAPI, NestJS
**AI & Data:** LangGraph, RAG, LLM evaluation, MCP, PostgreSQL, pgvector, Neo4j
**Cloud & Quality:** AWS Bedrock, Step Functions, GCP Cloud Run, Docker, pytest

## EDUCATION

**Arizona State University (ASU)** **08/2023 - 05/2025**
*M.S., Computer Science | GPA: 4.0* *Tempe, AZ, USA*

**Dhirubhai Ambani Institute of Information and Communication Technology** **08/2018 - 05/2022**
*B.Tech., Information and Communication Technology* *India*